	LINEAR PROGRAMMING	
	Convert the following LPP in the Standard form	
	Question	**Ans.**
1	*Minimise* $z=-3x_1+2x_2-x_3$ *subject to* $x_1-3x_2+2x_3\ge -6$ $3x_1+4x_3\le 3$ $-3x_1+5x_2\le 4$ $x_1, x_2\ge 0$, x_3 *unrestricted*	*Maximise* $z'=3x_1-2x_2+x_3'-x_3''+0s_1+0s_2+0s_3$ *subject to* $-x_1+3x_2-2x_3'+2x_3''+s_1=6$ $3x_1+4x_3'-4x_3''+s_2=3$ $-3x_1+5x_2+s_3=4$ $x_1, x_2, x_3', x_3'', s_1, s_2, s_3\ge 0$]
2	*Maximise* $z=3x_1+4x_2-2x_3$ *subject to* $6x_1-4x_2\le 5$ $3x_1+x_2+4x_3\ge 11$ $4x_1+3x_2\le 2$ $x_1, x_2\ge 0$	*Maximise* $z=3x_1+4x_2-2x_3'+2x_3''+0s_1+0s_2+0s_3$ *subject to* $6x_1-4x_2+s_1=5$ $3x_1+x_2+4x_3'-4x_3''-s_2=11$ $4x_1+3x_2+s_3=2$ $x_1, x_2, x_3', x_3'', s_1, s_2, s_3\ge 0$]
	Determine all basic solutions to the following problem	
	Question	**Ans.**
3	*Maximise* $z=x_1-2x_2+4x_3$ *subject to* $x_1+2x_2+3x_3=7$ $3x_1+4x_2+6x_3=15$	$x_1=1, x_2=3$ $x_1=1, x_3=2$ x_2, x_3 *unbounded solution*]
4	*Maximise* $z=x_1+3x_2+3x_3$ *subject to* $x_1+2x_2+3x_3=4$ $2x_1+3x_2+5x_3=7$	$x_1=2, x_2=1$ $x_1=1, x_3=1$ $x_2=-1, x_3=2$]
5	*Maximise* $z=2x_1-2x_2+4x_3-5x_4$ *subject to* $x_1+4x_2-2x_3+8x_4\le 2$ $-x_1+2x_2+3x_3+4x_4\le 1$ $x_1, x_2, x_3, x_4\ge 0$	$x_1=0, x_2=1/2,$ $x_1=0, x_4=1/4$ $x_1=8, x_3=3,$ $x_2, x_4=$ *unbounded* $x_2=1/2, x_3=0,$ $x_3=0, x_4=1/4$]
6	*Maximise* $z=x_1+x_2+3x_3$ *subject to* $x_1+2x_2+3x_3=9$ $3x_1+2x_2+2x_3=15$	$x_1=3, x_2=3$ $x_1=27/7, x_3=12/7$ $x_2=27/2, x_3=-6$]

	Solve the following linear programming problems by Simplex method	
7	*Maximise* $z=5x_1+4x_2$ *subject to* $6x_1+4x_2\le 24$ $x_1+2x_2\le 6$ $-x_1+x_2\le 1$ $x_2\le 2$ $x_1,x_2\ge 0$ [Ans: $x_1=3,\quad x_2=3/2\quad z_{\max}=21$]	8 *Maximise* $z=4x_1+10x_2$ *subject to* $2x_1+x_2\le 50$ $2x_1+5x_2\le 100$ $2x_1+3x_2\le 90$ $x_1,x_2\ge 0$ [Ans: $x_1=0,\quad x_2=20\quad z_{\max}=200$
9	*Maximise* $z=6x_1-2x_2+3x_3$ *subject to* $2x_1-x_2+2x_3\le 2$ $x_1+4x_3\le 4$ $x_1,x_2,x_3\ge 0$ Ans: $x_1=4,\ x_2=6,\ x_3=0\ \ z_{\max}=12$	10. *Maximise* $z=3x_1+2x_2$ *subject to* $x_1+x_2\le 4$ $x_1-x_2\le 2$ $x_1,x_2\ge 0$ [Ans: $x_1=3,\ x_2=1\ \ z_{\max}=11$]
11	*Maximise* $z=3x_1+2x_2+5x_3$ *subject to* $x_1+2x_2+x_3\le 430$ $3x_1+2x_3\le 460$ $x_1+4x_2\le 420$ $x_1,x_2,x_3\ge 0$ Ans: $x_1=0,\ x_2=100,\ x_3=230\ \ z_{\max}=1350$	12. *Minimise* $z=x_1-3x_2+3x_3$ *subject to* $3x_1-x_2+2x_3\le 7$ $2x_1-4x_2\ge -12$ $-4x_1+3x_2+8x_3\le 10$ $x_1,x_2,x_3\ge 0$ Ans: $x_1=4,\ x_2=5,\ x_3=0\ \ z_{\min}=-z'_{\max}=-11$]
13	*Maximise* $z=100x_1+50x_2+50x_3$ *subject to* $4x_1+3x_2+2x_3\le 10$ $3x_1+8x_2+x_3\le 8$ $4x_1+2x_2+x_3\le 6$ $x_1,x_2,x_3\ge 0$ Ans: $x_1=1/2,\ x_2=0,\ x_3=4\ \ z_{\max}=250$]	14. *Maximise* $z=3x_1+2x_2+5x_3$ *subject to* $x_1+x_2+x_3\le 9$ $2x_1+3x_2+5x_3\le 30$ $2x_1-x_2-x_3\le 8$ $x_1,x_2,x_3\ge 0$ Ans: $x_1=5,\ x_2=0,\ x_3=4,\ z_{\max}=3200/3$]

15	*Maximise* $z = 4x_1 + 3x_2 + 6x_3$ *subject to* $2x_1 + 3x_2 + 2x_3 \le 440$ $4x_1 + 3x_3 \le 470$ $2x_1 + 5x_2 \le 430$ $x_1, x_2, x_3 \ge 0$ Ans. $x_1 = 0,\ x_2 = 380/9,\ x_3 = 470/3$ $z_{\max} = 3200/3$]	16. *Maximise* $z = 4x_1 + 10x_2$ *subject to* $2x_1 + x_2 \le 10$ $2x_1 + 5x_2 \le 20$ $2x_1 + 3x_2 \le 18$ $x_1, x_2 \ge 0$ Ans $x_1 = 15/4,\ x_2 = 5/2,\ \ z_{\max} = 40$
17	*Maximise* $z = 4x_1 + x_2 + 3x_3 + 5x_4$ *subject to* $-4x_1 + 6x_2 + 5x_3 + 4x_4 \le 20$ $-3x_1 - 2x_2 + 4x_3 + x_4 \le 10$ $-8x_1 - 3x_2 + 3x_3 + 2x_4 \le 20$ $x_1, x_2, x_3, x_4 \ge 0$ Ans: *unbounded solution*	18. *Maximise* $z = 107x_1 + x_2 + 2x_3$ *subject to* $14x_1 + x_2 - 6x_3 + 3x_4 = 7$ $16x_1 + (1/2)x_2 - 6x_3 \le 5$ $3x_1 - x_2 - x_3 \le 0$ $x_1, x_2, x_3, x_4 \ge 0$ Ans: *unbounded solution*
19	*Maximise* $z = 100x_1 + 40x_2$ *subject to* $10x_1 + 4x_2 \le 2000$ $3x_1 + 2x_2 \le 900$ $6x_1 + 12x_2 \le 3000$ $x_1, x_2 \ge 0$ Has it an alternative optima? [Ans: $x_1 = 200,\ x_2 = 0,\ \ z_{\max} = 2000$ *alternate optima* $x_1 = 125,\ x_2 = 187.5$]	20 *Maximise* $z = 3x_1 + 5x_2 + 4x_3$ *subject to* $2x_1 + 3x_2 \le 8$ $2x_2 + 5x_3 \le 10$ $3x_1 + 2x_2 + 4x_3 \le 15$ $x_1, x_2, x_3 \ge 0$ Ans. $x_1 = 89/41,\ x_2 = 50/41,\ x_3 = 62/41$ $z_{\max} = 765/41$]
	Solve the following linear programming problems by Penalty (Big-M) method	
21	*Maximise* $z = 3x_1 - x_2$ *subject to* $2x_1 + x_2 \ge 2$ $x_1 + 3x_2 \le 3$ $x_2 \le 4$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 3,\ \ x_2 = 0\ \ z_{\max} = 9$]	22 *Maximise* $z = 3x_1 - x_2$ *subject to* $2x_1 + x_2 \le 2$ $x_1 + 3x_2 \ge 3$ $x_2 \le 4$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 3/5,\ \ x_2 = 4/5\ \ z_{\max} = 1$

23	$Maximise \quad z = 5x_1 - 2x_2 + 3x_3$ $subject\ to \quad 2x_1 + 2x_2 - x_3 \geq 2$ $3x_1 - 4x_2 \leq 3$ $x_2 + 3x_3 \leq 5$ $x_1, x_2, x_3 \geq 0$ Ans: $x_1 = 23/3,\ x_2 = 5,\ x_3 = 0\ \ z_{max} = 85/3$	24 $Minimise \quad z = x_1 + 2x_2 + x_3$ $subject\ to \quad x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3 \leq 1$ $\frac{3}{2}x_1 + 2x_2 + x_3 \geq 8$ $x_1, x_2, x_3 \geq 0$ [Ans: $No\ \ feasible\ \ solution$
25	$Minimise \quad z = 4x_1 + x_2$ $subject\ to \quad 3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$ Ans: $x_1 = 2/5, x_2 = 9/5 \quad z_{min} = 17/5$]	26 $Maximise \quad z = 10x_1 + 3x_2$ $subject\ to \quad x_1 + 2x_2 \geq 3$ $x_1 + 4x_2 \geq 4$ $x_1, x_2 \geq 0$ Ans: $x_1 = 0, \quad x_2 = 3/2 \quad z_{min} = 9/2$
27	$Minimise \quad z = 2x_1 + x_2$ $subject\ to \quad 3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 3$ $x_1, x_2 \geq 0$ Ans: $x_1 = 3/5, \quad x_2 = 6/5 \quad z_{min} = 12/5$	28 $Minimise \quad z = 2x_1 + 3x_2$ $subject\ to \quad x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$ Ans: $x_1 = 4, \quad x_2 = 1 \quad z_{min} = 11$
29	$Minimise \quad z = x_1 - 3x_2 - 2x_3$ $subject\ to \quad 3x_1 - x_2 + 2x_3 \geq 7$ $-2x_1 + 4x_2 \leq 12$ $-4x_1 + 3x_2 + 8x_3 \leq 10$ $x_1, x_2, x_3 \geq 0$ [Ans: $x_1 = 78/25, \quad x_2 = 114/25, x_3 = 11/10$ $z_{min} = -319/25$]	30 $Maximise \quad z = 4x_1 + 5x_2 + 2x_3$ $subject\ to \quad 2x_1 + x_2 + x_3 \leq 10$ $x_1 + 3x_2 + x_3 \leq 12$ $x_1 + x_2 + x_3 = 6$ $x_1, x_2, x_3 \geq 0$ [Ans: $x_1 = 3,\ x_2 = 3,\ x_3 = 0\ \ z_{max} = 27$]
31	$Maximise \quad z = 6x_1 + 4x_2$ $subject\ to \quad 2x_1 + 3x_2 \leq 30$ $3x_1 + 2x_2 \leq 24$ $x_1 + x_2 \geq 3$ $x_1, x_2 \geq 0$ Is the solution unique? If not, find another solution.	

	If the requirement vector $\begin{bmatrix}30\\24\\3\end{bmatrix}$ is changed to $\begin{bmatrix}24\\30\\3\end{bmatrix}$ is the solution still optimal? Ans: $x_1 = 8,\ x_2 = 0,\quad z_{max} = 48$ The alternate optimal basic feasible solution is $x_1 = 12/5,\ x_2 = 42/5,\ z_{max} = 48$

	Construct the Dual of the following LPP	
	Question	**Ans.**
32	*Minimise* $z = x_2 + 3x_3$ *subject to* $2x_1 + x_2 \le 3$ $x_1 + 2x_2 + 6x_3 \ge 5$ $-x_1 + x_2 + 2x_3 = 2$ $x_1, x_2, x_3 \ge 0$	*Maximise* $w = -3y_1 + 5y + 2y_3$ *subject to* $-2y_1 + y_2 - y_3 \le 0$ $-y_1 + 2y_2 + y_3 \le 1$ $6y_2 + 2y_3 \le 3$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]
33	*Minimise* $z = 3x_1 - 2x_2 + x_3$ *subject to* $2x_1 - 3x_2 + x_3 \le 5$ $4x_1 - 2x_2 \ge 9$ $-8x_1 + 4x_2 + 3x_3 = 8$ $x_1, x_2 \ge 0,\ x_3$ *unrestricted*	*Maximise* $w = -5y_1 + 9y + 8y_3$ *subject to* $-2y_1 + 4y_2 - 8y_3 \le 3$: $3y_1 - 2y_2 + 4y_3 \le -2$ $-y_1 + 3y_3 = 1$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]
34	*Maximise* $z = 3x_1 + 17x_2 + 9x_3$ *subject to* $-x_2 + x_3 \ge 3$ $-3x_1 + 2x_3 \le 1$ $2x_1 + x_2 - 5x_3 = 1$ $x_1, x_2, x_3 \ge 0$	*Minimise* $w = -3y_1 + y_2 + y_3$ *subject to* $-y_1 - 3y_2 + 2y_3 \ge 3$ $y_1 + y_3 \ge 17$ $-y_1 + 2y_2 - 5y_3 \ge 19$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]
35	*Maximise* $z = 2x_1 - x_2 + 3x_3$ *subject to* $x_1 - 2x_2 + x_3 \ge 4$ $2x_1 + x_3 \le 10$ $x_1 + x_2 + 3x_3 = 20$ $x_1, x_3 \ge 0, x_2$ *unrestricted*	*Minimise* $w = -4y_1 + 10y_2 + 20y_3$ *subject to* $-y_1 + 2y_2 + y_3 \ge 2$ $2y_1 + y_3 = -1$ $-y_1 + y_2 + 3y_3 \ge 3$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]

36	*Minimise* $z = x_1 - 3x_2 - 2x_3$ *subject to* $3x_1 - x_2 + 2x_3 \le 7$ $2x_1 - 4x_2 \ge 12$ $-4x_1 + 3x_2 + 8x_3 = 10$ $x_1, x_2 \ge 0, x_3$ *unrestricted*	*Maximise* $w = -7y_1 + 12y_2 + 10y_3$ *subject to* $-3y_1 + 2y_2 - 4y_3 \le 1$ $y_1 - 4y_2 + 3y_3 \le -3$ $-2y_1 + 8y_3 = -2$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]
37	*Maximise* $z = 2x_1 + x_2 + x_3$ *subject to* $x_1 + x_2 + x_3 \ge 6$ $3x_1 - 2x_2 + 3x_3 = 3$ $-4x_1 + x_3 \le 10$ $x_1, x_3 \ge 0, x_2$ *unrestricted*	*Minimise* $w = -6y_1 + 3y_2 + 10y_3$ *subject to* $-y_1 + 3y_2 - 4y_3 \ge 2$ $-y_1 - 2y_2 = 1$ $-y_1 + 3y_2 + y_3 \ge 1$ $y_1, y_3 \ge 0,\ y_2$ *unrestricted*]
38	*Maximise* $z = 2x_1 + 9x_2 + 11x_3$ *subject to* $x_1 - x_2 + x_3 \ge 3$ $-3x_1 + 2x_3 \le 1$ $2x_1 + x_2 - 5x_3 = 1$ $x_1, x_2, x_3 \ge 0$	*Minimise* $w = -3y_1 + y_2 + y_3$ *subject to* $-y_1 - 3y_2 + 4y_3 \ge 2$ $y_1 + y_2 \ge 9$ $-y_1 + 2y_2 - 5y_3 \ge 11$ $y_1, y_2 \ge 0,\ y_3$ *unrestricted*]
	Using Duality Solve the following linear programming problem	
39	*Minimise* $z = 4x_1 + 3x_2 + 6x_3$ *subject to* $x_1 + x_3 \ge 2$ $x_2 + x_3 \ge 5$ $x_2 \le 4$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 0,\ x_2 = 3,\ x_3 = 2,\quad z_{min} = 21$	40 *Minimise* $z = 5x_1 + 8x_2$ *subject to* $x_1 + x_2 \le 2$ $x_1 + 2x_2 \ge 0$ $-x_1 + 4x_2 \le 1$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 0,\quad x_2 = 0\quad z_{min} = 0$]
41	*Minimise* $z = 430x_1 + 460x_2 + 420x_3$ *subject to* $x_1 + 3x_2 + 4x_3 \ge 3$ $2x_1 + 4x_3 \ge 2$ $x_1 + 2x_2 \ge 5$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 1,\ x_2 = 2,\ x_3 = 0,\quad z_{min} = 1350$]	42 *Maximise* $z = 2x_1 + x_2$ *subject to* $2x_1 - x_2 \le 2$ $x_1 + x_2 \le 4$ $x_1 \le 3$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 1,\ x_2 = 2,\quad z_{max} = 6$]

43	*Maximise* $z = 3x_1 + 2x_2$ *subject to* $2x_1 + x_2 \le 5$ $x_1 + x_2 \le 3$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 2,\ x_2 = 1,\quad z_{\max} = 8$]	44 *Minimise* $z = 4x_1 + 14x_2 + 3x_3$ *subject to* $-x_1 + 3x_2 + x_3 \ge 3$ $2x_1 + 2x_2 - x_3 \ge 2$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 0,\ x_2 = 1,\ x_3 = 0,\quad z_{\min} = 14$]
45	*Maximise* $z = 5x_1 - 2x_2 + 3x_3$ *subject to* $2x_1 + 2x_2 - x_3 \ge 2$ $3x_1 - 4x_2 \le 3$ $x_2 + 3x_3 \le 5$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 23/3,\ x_2 = 5,\ x_3 = 0,\quad z_{\max} = 85/3$	46 *Minimise* $z = 2x_1 + 2x_2$ *subject to* $2x_1 + 4x_2 \ge 1$ $x_1 + 2x_2 \ge 1$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 1/3,\ x_2 = 1/3,\quad z_{\min} = 4/3$]
	Using Dual simplex method Solve the following linear programming problem	
47	*Minimise* $z = 2x_1 + 2x_2 + 4x_3$ *subject to* $2x_1 + 3x_2 + 5x_3 \ge 2$ $3x_1 + x_2 + 7x_3 \le 3$ $x_1 + 4x_2 + 6x_3 \le 5$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 0,\ x_2 = 2/3,\ x_3 = 0,\quad z_{\min} = 4/3$	48 *Maximise* $z = -3x_1 - 2x_2$ *subject to* $x_1 + x_2 \ge 1$ $x_1 + x_2 \le 7$ $x_1 + 2x_2 \ge 10$ $x_2 \le 3$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 4,\quad x_2 = 3\quad z_{\max} = -18$
49	*Minimise* $z = 6x_1 + 3x_2 + 4x_3$ *subject to* $x_1 + 6x_2 + x_3 = 10$ $2x_1 + 3x_2 + x_3 = 15$ $x_1, x_2, x_3 \ge 0$ [Ans: $x_1 = 20/3,\ x_2 = 5/9,\ x_3 = 0,\quad z_{\min} = 125/3$	50 *Minimise* $z = 6x_1 + x_2$ *subject to* $2x_1 + x_2 \ge 3$ $x_1 - x_2 \ge 0$ $x_1, x_2 \ge 0$ [Ans: $x_1 = 1,\ x_2 = 1,\quad z_{\min} = 7$]
52	*Minimise* $z = 2x_1 + x_2$ *subject to* $3x_1 + x_2 \ge 3$ $4x_1 + 3x_2 \ge 6$ $x_1 + 2x_2 \le 3$ $x_1, x_2 \ge 0$	53

	[Ans: $x_1 = 3/5,\ x_2 = 6/5,\ x_3 = 0, z_{\min} = 12/5$	*Minimise* $z = 6x_1 + 7x_2 + 3x_3 + 5x_4$ *subject to* $2x_1 + 5x_2 + x_3 + x_4 \geq 8$ $x_2 + 5x_3 - 6x_4 \geq 10$ $5x_1 + 6x_2 - 3x_3 + 4x_4 \geq 12$ $x_1, x_2, x_3, x_4 \geq 0$ [Ans: $x_1 = 0,\ x_2 = 30/11,\ x_3 = 16/11, x_4 = 0$ $z_{\min} = 258/11$]
54	*Minimise* $z = 3x_1 + 2x_2 + x_3 + 4x_4$ *subject to* $2x_1 + 4x_2 + 5x_3 + x_4 \geq 10$ $3x_1 - x_2 + 7x_3 - 2x_4 \geq 2$ $5x_1 + 2x_2 + x_3 + 6x_4 \geq 15$ $x_1, x_2, x_3, x_4 \geq 0$	
	Using the method of Lagrange's multipliers, solve the following N.L.P.P.	
	Question	**Ans.**
55	*Optimise* $z = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3$ *subject to* $x_1 + x_2 + x_3 = 7$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 4,\ x_2 = 2,\ x_3 = 1,\quad z_{\min} = -35$]
56	*Optimise* $z = 2x_1^2 + 2x_2^2 + 2x_3^2 - 24x_1 - 8x_2 - 12x_3 + 260$ *subject to* $x_1 + x_2 + x_3 = 11$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 6,\ x_2 = 2,\ x_3 = 3,\quad z_{\min} = 162$]
57	*Optimise* $z = 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 - 100$ *subject to* $x_1 + x_2 + x_3 = 20$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 5,\ x_2 = 11,\ x_3 = 4,\quad z_{\min} = 281$] (M.U. 2004, 06,13)

58	*Optimise* $z = 12x_1 + 8x_2 + 6x_3 - x_1^2 - x_2^2 - x_3^2 - 23$ *subject to* $x_1 + x_2 + x_3 = 10$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 5,\ x_2 = 3,\ x_3 = 2,\quad z_{max} = 35$]
59	*Maximise* $z = 6x_1 + 8x_2 - x_1^2 - x_2^2$ *subject to* $4x_1 + 3x_2 = 16$ $3x_1 + 5x_2 = 15$ $x_1, x_2 \geq 0$	[Ans: $x_1 = 35/11,\ x_2 = 12/11,\quad z_{max} = 16.50$])
60	*Optimise* $z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$ *subject to* $x_1 + x_2 + x_3 = 15$ $2x_1 - 5x_2 + 2x_3 = 20$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 33/9,\ x_2 = 10/9, x_3 = 8$ $\lambda_1 = 40/9,\ \lambda_2 = 52/9\quad z_{min} = 820/9$]
61	*Optimise* $z = x_1^2 + x_2^2 + x_3^2$ *subject to* $x_1 + x_2 + 3x_3 = 2$ $5x_1 + 2x_2 + x_3 = 5$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 37/46,\ x_2 = 16/46, x_3 = 13/46$ $\lambda_1 = 2/23, \lambda_2 = 7/23\quad z_{min} = 0.8476$
62	*Optimise* $z = 2x_1^2 + 3x_2^2 + x_3^2$ *subject to* $x_1 + x_2 + 2x_3 = 13$ $2x_1 + x_2 + x_3 = 10$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 2,\ x_2 = 1, x_3 = 5, \lambda_1 = 4, \lambda_2 = 2$ $z_{min} = 36$]
63	*Optimise* $z = 4x_1^2 - x_2^2 - x_3^2 - 4x_1x_2$ *subject to* $x_1 + x_2 + x_3 = 15$ $2x_1 - x_2 + 2x_3 = 20$ $x_1, x_2, x_3 \geq 0$	[Ans: $x_1 = 5095,\ x_2 = 3033, x_3 = 5.71$ $z_{min} = 83.87$]
	Using the Kuhn-Tucker conditions to solve the following N.L.P.P.	
64	*Maximise* $z = 2x_1^2 - 7x_2^2 + 12x_1x_2$ *subject to* $2x_1 + 5x_2 \leq 98$ $x_1, x_2 \geq 0$	[Ans: $x_1 = 44,\ x_2 = 2,\ z_{max} = 4900$]

65	*Maximise* $z = 8x_1 + 10x_2 - x_1^2 - x_2^2$ *subject to* $3x_1 + 2x_2 \le 6$ $x_1, x_2 \ge 0$	[Ans: $x_1 = 4/13,\ x_2 = 33/13,\ z_{\max} = 21.3$]
66	*Maximise* $z = 2x_1 + x_2 - x_1^2$ *subject to* $2x_1 + 3x_2 \le 6$ $2x_1 + x_2 \le 4$ $x_1, x_2 \ge 0$	[Ans: $x_1 = 2/3,\ x_2 = 14/9,\ z_{\max} = 22/9$]
67	*Maximise* $z = 2x_1 + 3x_2 - x_1^2 - 2x_2^2$ *subject to* $2x_1 + 3x_2 \le 6$ $5x_1 + 2x_2 \le 10$ $x_1, x_2 \ge 0$	[Ans: $x_1 =$, $x_2 =$, $z_{\max} =$]
68	*Maximise* $z = 4x_1 + 6x_2 - x_1^2 - x_2^2 - x_3^2$ *subject to* $x_1 + x_2 \le 2$ $2x_1 + 3x_2 \le 12$ $x_1, x_2, x_3 \ge 0$	[Ans: $x_1 = 1/2,\ x_2 = 3/2, x_3 = 0,\ z_{\max} = 17/2$]
69	*Maximise* $z = 2x_1 + 3x_2 - x_1^2 - x_2^2$ *subject to* $x_1 + x_2 \le 1$ $2x_1 + 3x_2 \le 6$ $x_1, x_2 \ge 0$	[Ans: $x_1 = 1/4,\ x_2 = 3/4,\ \lambda_1 = 3/2, \lambda_2 = 0$ $z_{\max} = 17/8$]
70	*Minimise* $z = 7x_1^2 + 5x_2^2 - 6x_1$ *subject to* $x_1 + 2x_2 \le 10$ $x_1 + 3x_2 \le 9$ $x_1, x_2 \ge 0$	[Ans:
	solve the following N.L.P.P.	
71	Find the relative maximum or minimum of the function $z = x_1^2 + x_2^2 + x_3^2 - 6x_1 - 10x_2 - 14x_3 + 103$	X1=3, x2=5, x3=7, zmax=20
72	Find the relative maximum or minimum of the funct $z = x_1^2 + x_2^2 + x_3^2 - 8x_1 - 10x_2 - 12x_3 + 100$	X1=4, x2=5, x3=6, zmax=23
73	Obtain the relative maximum or minimum, if any, of the function $z = 2x_1 + 6x_3 + 9x_2x_3 - 4x_1^2 - 9x_2^2 - 9x_3^2$	X1=1/4, x2=2/9, x3=4/9, zmax=19/12
74	Obtain the relative maximum or minimum, if any, of the function $z = 2x_1 + x_3 + 3x_2x_3 - x_1^2 - 3x_2^2 - 3x_3^2 + 17$	X1=1, x2=1/9, x3=2/9, zmax=18